

7 FORMULARIUM

Note: numbers in the right column refer to equations in the textbook. In some cases, equations are not explicitly given in the textbook, but are part of the text or a footnote. In those cases, they are numbered according to the closest numbered equation.

7.1 GASLAWS, WATER VAPOUR AND THERMODYNAMICS (APPENDIX B)

(T in Kelvin, unless otherwise stated; some definitions are given in table B.2: they are referred to as TB2.x)

Equation of state	$p = \rho \frac{R^*}{M} T = \rho R T$	B.1					
Hydrostatic equilibrium	$dp = -g\rho dz$ $g = 9.8 \text{ m s}^{-2}$	B.11					
Thermodynamics	$dQ = c_v dT + p d\alpha$ $dQ = c_p dT - \frac{dp}{\rho}$ $c_p - c_v = R$	B.10 B.7					
Dry air (occurs in text)	$M_d = 29 \text{ kg kmol}^{-1}$	B.3a					
Dry-adiabatic	$\frac{dT}{T} = \frac{R}{c_p} \frac{dp}{p}$ $\theta = T \left(\frac{p_0}{p} \right)^{\frac{R}{c_p}}$	B.13 B.14					
Relative humidity RH	$RH = \frac{e}{e_{\text{sat}}}$	TB2.6					
Specific humidity q	$q = \frac{\rho_v}{\rho}$ $q \approx \frac{R_d}{R_v} \frac{e}{p}$ $q = \frac{r}{1+r}$	TB2.4 B.19 B.17					
Mixing ratio r	$r = \frac{\rho_v}{\rho_d}$	TB.3					
Dew point T_d	$e = e_{\text{sat}}(T_d)$	TB2.7					
Virtual temperature T_v	$T_v = \frac{T}{1 - \left(1 - \frac{R_d}{R_v}\right) \frac{e}{p}} \approx T(1 + 0.61q)$	TB2.5					
Wet bulb temperature T_w and psychrometer-equation	$e = e_{\text{sat}}(T_w) - \frac{R_v}{R_d} \frac{c_p}{L_v} p(T - T_w) = e_{\text{sat}}(T_w) - \gamma(T - T_w)$ with $\gamma = \frac{R_v}{R_d} \frac{c_p}{L_v} p$	TB2.8					
Thermodynamic properties of dry and moist air (note: L_v is given in the text, section B.5)		depends on					
	property	T	q	p	value	unit	
	R^*				8314	$\text{J kmol}^{-1}\text{K}^{-1}$	B.1a
	R_v				462	$\text{J kg}^{-1}\text{K}^{-1}$	B.4a
	R_d				287	$\text{J kg}^{-1}\text{K}^{-1}$	B.3b
	R		•		$287(1 + 0.61q)$	$\text{J kg}^{-1}\text{K}^{-1}$	B.6
	c_{pd}				1004	$\text{J kg}^{-1}\text{K}^{-1}$	B.7a
	c_{vd}				717	$\text{J kg}^{-1}\text{K}^{-1}$	B.7b
	c_p		•		$1004(1 + 0.84q)$	$\text{J kg}^{-1}\text{K}^{-1}$	B.8
	c_v		•		$717(1 + 0.37q)$	$\text{J kg}^{-1}\text{K}^{-1}$	B.9
	L_v	•			$2501000(1 - 0.00095(T - 273.15))$	J kg^{-1}	B.23b
	γ	•	•	•	$65.5 \frac{1 + 0.84q}{1 - 0.00095(T - 273.15)} \frac{p}{101300}$	Pa K^{-1}	B.23
	e_{sat} (over water)	•			$611.2 \exp \left[\frac{17.62(T - 273.15)}{-30.03 + T} \right]$	Pa	B.20a

	s (over water)	•		$e_{\text{sat}}(T) \frac{4284}{(-30.03+T)^2}$ (with e_{sat} over water)	Pa K ⁻¹	B.21a
	e_{sat} (over ice)	•		$611.2 \exp \left[\frac{22.46(T - 273.15)}{-0.53 + T} \right]$	Pa	B.20b
	s (over ice)	•		$e_{\text{sat}}(T) \frac{6123}{(-0.53+T)^2}$ (with e_{sat} over ice)	Pa K ⁻¹	B.21b

7.2 RADIATION (PARTLY FROM APPENDIX A) AND SOIL HEAT FLUX

Planck curve	$M_{b\lambda} = \frac{C_1 \lambda^{-5}}{\left[\exp\left(\frac{C_2}{\lambda T}\right) - 1 \right]}$ W m ⁻² μm ⁻¹ $C_1 = 3.74 \cdot 10^8$ W m ⁻² μm ⁴ , $C_2 = 1.439 \cdot 10^4$ μm K	A.1
Stefan-Boltzmann	$M_b = \sigma T^4$ W m ⁻² with $\sigma = 5.67 \cdot 10^{-8}$ W m ⁻² K ⁻⁴	A.2
Wien	$\lambda_m T = 2897.8$ μm K	A.3
Lambert-Beer	$dI_\lambda = -I_\lambda k_\lambda q \rho ds$	2.9
Radiation at top of atmosphere	$K_0^\downarrow = \bar{I}_0 \left(\frac{\bar{d}_{\text{Sun}}}{d_{\text{Sun}}} \right)^2 \cos(\theta_z)$ $\bar{I}_0 = 1365$ W m ⁻²	2.3
	$\cos(q_z) = \sin(\delta) \sin(\phi) + \cos(\delta) \cos(\phi) \cos(\omega)$	A.10
	$\overline{K_0^\downarrow} = \frac{\bar{I}_0}{\pi} \left(\frac{\bar{d}_{\text{Sun}}}{d_{\text{Sun}}} \right)^2 [\omega_s \sin(\delta) \sin(\phi) + \cos(\delta) \cos(\phi) \sin(\omega_s)]$ (θ_z : zenith angle, δ : declination angle, ϕ : latitude, ω : hour angle, ω_s : daylength (radians))	A.15
	$\overline{K^\downarrow} = \overline{K_0^\downarrow} \left(a + b \frac{n}{N_d} \right)$ (N_d : daylength (hours) = $2 \frac{24}{2\pi} \omega_s$)	2.17
Global radiation (empirical estimate daily sum)		
Extinction of radiation	$I_\lambda = I_{\lambda 0} \exp \left[-m_r \int_0^\infty k_{\lambda,i} q_i \rho dz \right] = I_{\lambda 0} \tau_{\lambda\theta,i}$	2.10
Global radiation as sum of direct and diffuse	$K^\downarrow = S + D = I \cos(\theta_z) + D$	2.4
Optical thickness	$\delta_{\lambda,i} \equiv \int_0^\infty k_{\lambda,i} q_i \rho dz$	2.11
Optical mass (vertical, relative)	$m_a = \int_\infty^0 \rho ds$	2.5
	$m_v = \int_\infty^0 \rho dz$	2.6
	$m_a = m_v m_r$	2.7
	$m_r \approx [\cos(\theta_z)]^{-1}$	2.8
Transmissivities	$\tau_{b,\theta} = \frac{I}{I_0}$	2.15
	$\tau_b = \frac{K^\downarrow}{K_0^\downarrow}$	2.16
Models for broadband transmissivities	$\tau_{b,\theta} = \left[(\tau_{bv,cda})^{m_r} \right]^{T_L} = \exp(\delta_{b,cda} m_r T_L)$	2.18
	$\frac{K^\downarrow}{K_0^\downarrow} = \tau_b = 0.868 e^{-0.0387 m_r T_{L,2}}$ (with T_L the Linke turbidity and $\delta_{b,cda}$ the broadband optical thickness of a clean, dry atmosphere).	2.19
Brunt	$\varepsilon_{a,\text{clear}} = c_1 + c_2 \sqrt{e_a}$ with e_a in mbar and $c_1 = 0.52$ and $c_2 = 0.065$ hPa ^{-0.5}	2.25
Incoming radiation longwave (empirical estimate)	$L^\downarrow = \varepsilon_a \sigma T_a^4$ W m ⁻² , with	2.24
	$\varepsilon_a = f_{\text{cloud}} + (1 - f_{\text{cloud}}) \varepsilon_{a,\text{clear}}$	2.26

Upwelling radiation	longwave	$L^\uparrow = L_e^\uparrow + (1 - \varepsilon_s)L^\downarrow$	2.28
		with $L_e^\uparrow \approx \varepsilon_s \sigma T_s^4$	2.27
Soil heat flux		$G = -\lambda_s \frac{\partial T}{\partial z_d}$	2.29
		$\frac{\partial G}{\partial z_d} = -\rho_s c_s \frac{\partial T}{\partial t}$	2.30
		$\frac{\partial T}{\partial t} = \kappa_s \frac{\partial^2 T}{\partial z_d^2}$	2.31
Sine-wave forcing of T at surface		$T(z_d, t) = \bar{T} + A(0)e^{-\frac{z_d}{D}} \sin\left(\omega t - \frac{z_d}{D}\right)$ with $\omega = \frac{2\pi}{P}$ (P is period of sine wave)	2.35
		Damping depth (penetration depth): $D = \sqrt{\frac{2\lambda}{\omega \rho_s c_s}} = \sqrt{\frac{2\kappa_s}{\omega}}$	2.37
		Amplitude at depth z_d : $A(z_d) = A(0)e^{-\frac{z_d}{D}}$	2.36
		Flux density at depth z_d and time t : $G(z_d, t) = A(0)e^{-\frac{z_d}{D}} \sqrt{\omega \rho_s c_s \lambda_s} \sin\left[\omega t - \frac{z_d}{D} + \frac{\pi}{4}\right]$	2.38
Force-restore method		$\frac{\partial T_{top}}{\partial t} = \frac{1}{c_s d_{top}} \left\{ \underbrace{(Q^* - H - L_v E)}_{\text{force}} - \underbrace{\Lambda_s (T_{top} - T_{bot})}_{\text{restore}} \right\}$ with: $\Lambda_s = C_s d_{top} \omega$ and $d_{top} = \sqrt{\kappa_s / 2\omega}$	2.40
Calorimetric method		$\frac{\partial T_G}{\partial t} \approx \frac{1}{c_s z_m} (G_0 - G_m)$ where subscript 0 indicates the surface, subscript m indicates location of flux measurement and T_G is the mean temperature between surface and measurement location. (equation is part of text, above equation 2.44; see also equation 3.16)	2.44
Frost penetration depth		$z_f = a\sqrt{-I_n}$	2.51

7.3 TURBULENT FLUXES

K-theory (equation 3.1 in the book is the example for temperature)	$H = -\bar{\rho} c_p K_h \frac{\partial \bar{\theta}}{\partial z}$	3.1a
	$E = -\bar{\rho} K_e \frac{\partial \bar{q}}{\partial z}$	3.1b
	$\tau = \bar{\rho} K_m \frac{\partial \bar{u}}{\partial z}$	3.1c
Flux densities (do not occur as numbered equations in the book, see page 88)	$H = \bar{\rho} c_p \overline{w'\theta'}$	3.13a
	$E = \bar{\rho} \overline{w'q'}$	3.13b
	$\tau = -\bar{\rho} \overline{u'w'}$	3.13c
Heat flux divergence	$\frac{\partial \bar{\theta}}{\partial t} = -\frac{1}{\bar{\rho} c_p} \frac{\partial H}{\partial z} \Rightarrow \int_0^z \bar{\rho} c_p \frac{\partial \bar{\theta}}{\partial t} dz = -[H_z - H]$	3.16
Turbulent scales (moisture, temperature, velocity (friction velocity), virtual temperature)	$u_* = \sqrt{\frac{\tau}{\bar{\rho}}}$	3.17a
	$\theta_* = -\frac{H}{\bar{\rho} c_p u_*}$	3.17b
	$q_* = -\frac{E}{\bar{\rho} u_*}$	3.17c
	$\theta_{v*} = -\frac{H_v}{\bar{\rho} c_p u_*}$	3.17d
TKE and TKE equation	$\bar{e} = \frac{1}{2} (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$	3.9

	$\frac{d\bar{e}}{dt} = -\overline{u'w'} \frac{\partial \bar{u}}{\partial z} + g \frac{\overline{w'\theta_v'}}{\theta_v} - \varepsilon + \dots$	3.10
Stability parameter and Obukhov length	Stability parameter $\frac{z}{L} = -\frac{\kappa z g}{\bar{\theta}_v} \frac{H_v}{\bar{\rho} c_p u_*^3}$	3.19
	Obukhov length: $L = \frac{\bar{\theta}_v}{\kappa g} \frac{u_*^2}{\theta_{v*}} = -\frac{\bar{\rho} c_p \bar{\theta}_v u_*^3}{\kappa g H_v}$	3.19b
Aerodynamic resistance	$H = -\bar{\rho} c_p \frac{\bar{\theta}(z_2) - \bar{\theta}(z_1)}{r_{ah}}$	3.24a
	$E = -\bar{\rho} \frac{\bar{q}(z_2) - \bar{q}(z_1)}{r_{ae}}$	3.24b
	for neutral conditions: $r_{ae} = r_{ah} = r_{am} = \frac{\ln\left[\frac{z_2}{z_1}\right]}{\kappa u_*}$ (with u_*)	3.25a
	$r_{ah} = \frac{\ln\left[\frac{z_{\theta 2}}{z_{\theta 1}}\right] \ln\left[\frac{z_{u2}}{z_{u1}}\right]}{\kappa^2 [\bar{u}(z_2) - \bar{u}(z_1)]}$ (with windspeed) ($\kappa=0.4$, von Karman constant)	3.25b
Richardson numbers	$Ri_f = \frac{g}{\theta_v} \frac{\overline{w'\theta_v'}}{\overline{u'w'} \frac{\partial \bar{u}}{\partial z}}$	3.11
	$Ri_g = \frac{g}{\theta_v} \frac{\frac{\partial \bar{\theta}_v}{\partial z}}{\left[\frac{\partial \bar{u}}{\partial z}\right]^2}$	3.33
	$Ri_f = \frac{K_h}{K_m} Ri_g$	3.34a
Buoyancy flux (these are hidden in footnote 6 (page 85) and footnote 16 (page 101))	Surface flux (energy): $H_v = H(1 + 0.61\bar{q}) + 0.61c_p\bar{\theta}E$	3.10b
	Turbulent flux (kinematic): $\overline{w'\theta_v'} = \overline{w'\theta'}(1 + 0.61\bar{q}) + 0.61\bar{\theta} \overline{w'q'}$	3.19b
Flux-gradient relationships	$\frac{\partial \bar{\theta}}{\partial z} \frac{\kappa z}{\theta_*} = \phi_h \left(\frac{z}{L}\right)$	3.20a
	$\frac{\partial \bar{q}}{\partial z} \frac{\kappa z}{q_*} = \phi_e \left(\frac{z}{L}\right)$	3.20b
	$\frac{\partial \bar{u}}{\partial z} \frac{\kappa z}{u_*} = \phi_m \left(\frac{z}{L}\right)$	3.20c
	$Ri_g = \frac{\phi_h}{\phi_m^2} \frac{z}{L}$	3.34b
	$\phi_h = \phi_e = \phi_m^2 = \left(1 - 16 \frac{z}{L}\right)^{-\frac{1}{2}}$ (if $L < 0$)	3.21a
	$\phi_h = \phi_e = \phi_m = \left(1 + 5 \frac{z}{L}\right)$ (if $L > 0$)	3.21b
	With the above functions: $\frac{z}{L} = Ri_g$ (if $L < 0$)	3.36a
	$\frac{z}{L} = \frac{Ri_g}{1 - 5Ri_g}$ (if $L > 0$)	3.36b
	Neutral conditions: $\frac{\partial \bar{\theta}}{\partial z} \frac{\kappa z}{\theta_*} = 1$	3.20d
	$\frac{\partial \bar{q}}{\partial z} \frac{\kappa z}{q_*} = 1$	3.20e
	$\frac{\partial \bar{u}}{\partial z} \frac{\kappa z}{u_*} = 1$ (these are not explicitly included in the book, but the consequence of 3.20)	3.20f
Integrated flux-gradient relationships	$\bar{\theta}(z_{\theta 2}) - \bar{\theta}(z_{\theta 1}) = \frac{\theta_*}{\kappa} \left[\ln\left(\frac{z_{\theta 2}}{z_{\theta 1}}\right) - \Psi_h\left(\frac{z_{\theta 2}}{L}\right) + \Psi_h\left(\frac{z_{\theta 1}}{L}\right) \right]$	3.29a
	$\bar{u}(z_{u2}) - \bar{u}(z_{u1}) = \frac{u_*}{\kappa} \left[\ln\left(\frac{z_{u2}}{z_{u1}}\right) - \Psi_m\left(\frac{z_{u2}}{L}\right) + \Psi_m\left(\frac{z_{u1}}{L}\right) \right]$	3.29b

	$\Psi_m\left(\frac{z}{L}\right) = 2 \ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2 \arctan(x) + \frac{\pi}{2} \quad (\text{if } L < 0)$ $\text{with } x = \left(1 - 16\frac{z}{L}\right)^{\frac{1}{4}}$	3.30a
	$\Psi_h\left(\frac{z}{L}\right) = 2 \ln\left(\frac{1+x^2}{2}\right) \quad (\text{if } L < 0)$ $\text{with } x = \left(1 - 16\frac{z}{L}\right)^{\frac{1}{4}}$	3.30b
	$\Psi_m\left(\frac{z}{L}\right) = \Psi_h\left(\frac{z}{L}\right) = -5\frac{z}{L} \quad (\text{if } L > 0)$	3.30c
	Neutral: $\bar{u}(z_{u2}) - \bar{u}(z_{u1}) = \frac{u_*}{\kappa} \ln\left(\frac{z_{u2}}{z_{u1}}\right)$ $\bar{\theta}(z_{\theta2}) - \bar{\theta}(z_{\theta1}) = \frac{\theta_*}{\kappa} \ln\left(\frac{z_{\theta2}}{z_{\theta1}}\right)$	3.23a 3.23b
General equations for H and u_* (these equations do not occur explicitly in the book, but are directly related to 3.43)	$H = \frac{-\bar{\rho}c_p\kappa^2\Delta\bar{\theta}\Delta\bar{u}}{\left[\ln\left(\frac{z_{\theta2}}{z_{\theta1}}\right) - \Psi_h\left(\frac{z_{\theta2}}{L}\right) + \Psi_h\left(\frac{z_{\theta1}}{L}\right)\right]\left[\ln\left(\frac{z_{u2}}{z_{u1}}\right) - \Psi_m\left(\frac{z_{u2}}{L}\right) + \Psi_m\left(\frac{z_{u1}}{L}\right)\right]}$	3.43a
	$u_* = \frac{\kappa\Delta\bar{u}}{\left[\ln\left(\frac{z_{u2}}{z_{u1}}\right) - \Psi_m\left(\frac{z_{u2}}{L}\right) + \Psi_m\left(\frac{z_{u1}}{L}\right)\right]}$	3.43b
Analytical solutions ($L < 0$)	$H = \frac{-\bar{\rho}c_p\kappa^2\Delta\bar{\theta}\Delta\bar{u}}{\left[\ln\left(\frac{z_2}{z_1}\right)\right]^2} (1 - 16Ri_b)^{\frac{3}{4}}$	3.47
	$u_* = \frac{\kappa\Delta\bar{u}}{\left[\ln\left(\frac{z_2}{z_1}\right)\right]} (1 - 16Ri_b)^{\frac{1}{4}}$	3.48
	with: $Ri_b = \sqrt{z_1 z_2} \left[\ln\left(\frac{z_2}{z_1}\right)\right] \frac{g}{\theta_v} \frac{\Delta\bar{\theta}_v}{(\Delta\bar{u})^2}$	3.49
	(as a consequence: $L = \frac{\theta_v}{g} \frac{(\Delta\bar{u})^2}{\Delta\bar{\theta}_v \ln\left(\frac{z_2}{z_1}\right)}$)	3.49b
Analytical solutions ($L > 0$)	$H = \frac{-\bar{\rho}c_p\kappa^2\Delta\bar{\theta}\Delta\bar{u}}{\left[\ln\left(\frac{z_2}{z_1}\right)\right]^2} (1 - 5Ri_b)^2 \quad (\text{provided that } Ri_b < 0.2)$	3.50
	$u_* = \frac{\kappa\Delta\bar{u}}{\left[\ln\left(\frac{z_2}{z_1}\right)\right]} (1 - 5Ri_b) \quad (\text{provided that } Ri_b < 0.2)$	3.51
	with: $Ri_b = (z_2 - z_1) \frac{g}{\theta_v} \frac{\Delta\bar{\theta}_v}{(\Delta\bar{u})^2}$	3.52
	(as a consequence: $L = \frac{(z_2 - z_1)(1 - 5Ri_b)}{\ln\left(\frac{z_2}{z_1}\right) Ri_b}$)	3.52b

7.4 SOIL WATER FLOW

Warrilow model	$\frac{d\theta}{dt} = \frac{1}{D_r} [P - E(\theta) - D(\theta)]$	4.4
	$E(\theta) = \beta_w(\theta)E_p \quad \text{with} \quad \beta_w = \begin{cases} \frac{1}{\theta_c - \theta_w} & \text{for } \theta_c < \theta < \theta_s \\ \frac{\theta - \theta_w}{\theta_c - \theta_w} & \text{for } \theta_w < \theta < \theta_c \\ 0 & \text{for } \theta < \theta_w \end{cases}$	4.2
	$D = k(\theta) = k_s \left(\frac{\theta - \theta_w}{\theta_s - \theta_w}\right)^n$	4.3
Vapour pressure	$\ln\left(\frac{e}{e_{\text{sat}}}\right) = 7.5 \times 10^{-7} \text{ cm}^{-1} (h + \pi)$	4.10
Effective conductivity	$k_{\text{eff}} = \frac{\sum_{j=1}^N L_j}{\sum_{j=1}^N \frac{L_j}{k_j}}$	4.15
Continuity eq.	$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z} - S$	4.21

Richards' equation	$\frac{\partial \theta}{\partial t} = C(h) \frac{\partial h}{\partial t} = \frac{\partial \left[k(h) \left(\frac{\partial h}{\partial z} + 1 \right) \right]}{\partial z} - S(z)$	4.22
Soil hydraulic functions	$\theta(h) = \theta_r + \frac{\theta_s - \theta_r}{(1 + \alpha h ^n)^{\frac{n-1}{n}}}$	4.23
	$k(\theta) = k_s S_e^\lambda \left[1 - \left(1 - S_e^{\frac{n}{n-1}} \right)^{\frac{n-1}{n}} \right]^2$	4.24
	with $S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}$	4.25
Infiltration	Horton: $I = I_f + (I_0 - I_f)e^{-\beta t}$	4.26
	$I_{cum} = I_f t + \frac{I_0 - I_f}{\beta} [1 - e^{-\beta t}]$	4.27
	Green Ampt horizontal: $I_{cum} = [-2k_t h_f (\theta_t - \theta_i)]^{\frac{1}{2}} t^{\frac{1}{2}} = S t^{\frac{1}{2}}$	4.35
	Green Ampt vertical: $t = \frac{\theta_t - \theta_i}{k_t} \left[S_f + h_f \ln \left(\frac{S_f - h_f}{-h_f} \right) \right]$	4.40
Capillary rise	$z_c = \frac{2\sigma \cos \varphi}{\rho g r}$	4.7
	$Z = \int_0^z dz = - \int_0^h \frac{k(h)}{q + k(h)} dh$	4.43

7.5 SOLUTE TRANSPORT

Solute flux	$J = J_{dif} + J_{con} + J_{dis} = qC_l - \theta(D_{dif} + D_{dis}) \frac{\partial C_l}{\partial z}$	5.5
Mass balance	$\frac{\partial C_T}{\partial t} = - \frac{\partial J}{\partial z} - S_s$	5.6
General transport equation	$\frac{\partial}{\partial t} (\rho_b C_a + \theta C_l) = \frac{\partial}{\partial z} \left(\theta D_e \frac{\partial C_l}{\partial z} \right) - \frac{\partial}{\partial z} (q C_l) - S_s$	5.8
Pulse breakthrough	$C(z, t) = \frac{-zC_0}{2\sqrt{\pi D_e t^3}} \exp \left(- \frac{(z - vt)^2}{4D_e t} \right)$	5.12
Retardation factor	$R = 1 + \frac{\rho_b S_d}{\theta}$	5.16a
First order decay	$M(t) = M_0 e^{-\mu t}$	5.20
Salinization root zones	$C_l(z) = \frac{C_0}{1 + (L_f - 1) \frac{ z }{D_r}}$	5.26
	$\frac{\bar{C}}{C_0} = \frac{1}{1 - L_f} \ln \left(\frac{1}{L_f} \right)$	5.30
Residence time in groundwater	$T_{res} = \frac{\phi H}{R} \ln \left(\frac{L}{2x} \right)$	5.32
	$C_{out} = C_{in} + (C_{orig} - C_{in}) e^{\frac{-Rt}{\phi H}}$	5.37

7.6 PLANT TRANSPORT PROCESSES

Root water uptake (macroscopic)	$S_p(z) = \frac{L_{root}(z)}{\int_{-D_r}^0 L_{root}(z) dz} T_p$	6.15
	$S(z) = \alpha_{rw}(z) \alpha_{rs}(z) S_p(z) \quad \text{with} \quad T_a = \int_{-D_r}^0 S(z) dz$	6.17
		6.18
Flow in vessels	$v = - \frac{r^2}{8\eta} \frac{\partial H_p}{\partial x}$	6.19

Fluxes at stomate level	Transpiration: $E = -\rho \frac{q_e - q_i}{r_s}$	6.21
	Photosynthesis and net CO ₂ exchange: $A_n = -F_c = \rho \frac{q_{ce} - q_{ci}}{r_{s,c}} = \rho \frac{q_{ce} - q_{ci}}{1.6r_s} = \frac{g_s}{1.6} \rho (q_{ce} - q_{ci})$	6.23
Empirical dependence of internal CO ₂ concentration (q_{ci}) on external vapour pressure deficit (D_e)	$\frac{q_{ci} - \Gamma}{q_{ce} - \Gamma} = f_{max} - a_d D_e$ with $f_{max} \approx 0.9$ and $a_d \approx 0.07 \text{ (kPa)}^{-1}$ (C ₃ plants) or $a_d \approx 0.15 \text{ (kPa)}^{-1}$ (C ₄ plants)	6.28
Reaction of stomatal conductance (for CO ₂) on assimilation and transpiration	$g_{s,c} \approx \frac{a_1}{\rho(q_{ce} - \Gamma)} A_n - \frac{a_2}{1.6\rho a_3 D_0} T$ where a_1, a_2, a_3 and D_0 are plant-dependent constants (some of which are not dimensionless)	6.30
Ecosystem level CO ₂ exchange	$NEE = A_n - R_s = A_g - R_d - R_s$	6.32
Crop yield (dry matter production DM_a), water productivity (WP_T)	$WP_T = \frac{DM_a}{T_a} = 1000 \frac{\mu}{D_a}$ (note that the factor 1000 has units of kg m ⁻³ ; this implies that DM_a is in kg m ⁻² and T_a in m; μ and D_a should have identical units, usually Pa)	6.35
Leaf temperature	$T_{leaf} = T_a + \frac{\tau_b}{\rho c_p} (Q_{leaf}^* - L_v E_{leaf})$	6.41

7.7 COMBINATION METHODS

Energy balance method	$\beta \equiv \frac{H}{L_v E} = \frac{c_p \Delta \bar{\theta}}{L_v \Delta \bar{q}} = \gamma \frac{\Delta \bar{\theta}}{\Delta \bar{e}}$	7.1
	$H = \beta \frac{Q^* - G}{1 + \beta}$	7.2a
	$L_v E = \frac{Q^* - G}{1 + \beta}$	7.2b
Penman or Penman-Monteith	$L_v E = \frac{s(Q^* - G) + \frac{\rho c_p}{r_a} [e_{sat}(\bar{T}_a) - \bar{e}_a]}{s + \gamma}$	7.13
	$L_v E = \frac{s(Q^* - G) + \frac{\rho c_p}{r_a} [e_{sat}(\bar{T}_a) - \bar{e}_a]}{s + \gamma \left(1 + \frac{r_c}{r_a}\right)}$	7.16
Equilibrium evaporation and Priestley-Taylor	$L_v E_{eq} = \frac{s}{s + \gamma} (Q^* - G)$	7.17
	$L_v E = \alpha_{PT} \frac{s}{s + \gamma} (Q^* - G)$ with $\alpha_{PT} \approx 1.26$	7.18
Makkink	$L_v E = 0.65 \frac{s}{s + \gamma} K^\downarrow$	7.20